

CSES Problem Set

Counting Patterns

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Time limit: 1.00 s **Memory limit:** 512 MB

Given a string and patterns, count for each pattern the number of positions where it appears in the string.

Input

The first input line has a string of length n .

The next input line has an integer k : the number of patterns. Finally, there are k lines that describe the patterns.

The string and the patterns consist of characters a–z.

Output

For each pattern, print the number of positions.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq 5 \cdot 10^5$
- the total length of the patterns is at most $5 \cdot 10^5$

Example

Input:
aybabbtu
3
bab
abc
a

Output:
1
0
2